

Patrick M. TIMONS

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EDUCATION

Artificial Intelligence and Decision Making — *Bachelor of Science*
Massachusetts Institute of Technology

- *Software Engineering*: Software Performance Engineering, Algorithms II, Algorithms I, Computer Programming, Computation Structures, Low-Level Programming, Mathematics for Comp Sci
- *Mathematics and Machine Learning*: Inference and Information (G), Machine Learning (G), Machine Learning (UG), Deep Learning (G), Deep Learning (UG), Probability, Statistics, Linear Algebra and Optimization, Representation, Inference, and Reasoning in AI, Quantitative NLP, Computer Vision
- *Economics and Social Science*: Networks, Microeconomics, Macroeconomics, Information Policy
- *Other Science*: Linguistics, Chemistry, Biology

AUG 2021 - MAY 2025
GPA: 4.7/5.0

WORK EXPERIENCE

AWS Billing — *SDE*

- Utilized caching in an internal batch compute framework to launch Attribute Based Discounts in the ZHY partition
- Created infrastructure to support regression testing for strategic AWS customers undergoing major billing system changes
- Used Java, Batch Compute Framework, Protobuf, Apache Parquet, DuckDB, Build Tools, Guice, Spock, AWS CDK

SEP 2025 - PRESENT

Stealth Start-up — *Co-Founder, CTO*

- MIT \$100K Finalist: Recognized among top MIT startups for innovation and market potential
- Winner of Harvard Law Entrepreneurship Competition: Gained access to mentorship from 8 Harvard Law School attorneys
- MIT Sandbox Grant Recipient: Awarded \$10,000 (\$5,000 cash and \$5,000 AWS Credits) for product development and scaling

SEP 2024 - MAR 2025

AWS Billing — *SDE Intern*

- Designed and implemented an end-to-end data pipeline to process real-time billing data of 550,000 enterprise AWS customers
- Built an ML-powered forecasting service for AWS cost-savings data analysis, served through a query-based API and a front-end dashboard
- Used AWS, Athena, IaC, CDK, SageMaker, S3, API Gateway, Data Catalog, Java, TypeScript, Python

JUN 2024 - AUG 2024

TECHNICAL PROJECTS

Decaf Compiler

- Building a compiler for Decaf programming language from scratch with Java (WIP)

GitHub

Improving Deep Learning Based Molecular Fingerprints Through Informed Resampling

- Worked in a small team to research improvements to pretraining methods for transformer-based molecular encoders
- Created custom RoBERTa model through the use of transformers and bert-loves-chemistry (ChemBERTa) software packages with 9.1 percent improvement in Spearman’s rank correlation coefficient on downstream tasks compared to the base model
- Coauthored 5-page ACL-submission-style write-up about project and results

GitHub

Recovering Latent Variables with Variational Autoencoders despite Training Bias

- Researched how beta-regularization robustifies VAEs to training bias when attempting to recover latent variables
- Trained models with PyTorch Lightning and used scientific computing libraries to generate training data and visualize results

GitHub

EXTRACURRICULAR ACTIVITIES AND HONORS

MIT Men’s Lacrosse — *Team Member*

- Awarded USILA Academic All-American for the 2025 Season
- Received Ben Martin Award for “senior lacrosse player who has continuously exhibited the qualities of dedication, determination, and spirit”
- Voted Most Improved for 2023 season, NEWMAC All-Academic Award (2023, 2024, 2025), 2022 NEWMAC Champion

AUG 2021 - MAY 2025

Global Teaching Labs — *Applied Math Teacher*

- Taught selected topics in Operations Research with a focus on Bayesian Inference to high school students in Cremona, Italy

JAN 2024 - JAN 2024

SKILLS

Foundational Data Structures & Algorithms, Programming (Java, Python, Bash, C, TypeScript)

Design Patterns and Programming Paradigms Model-first development, Dependency Injection, Object-Oriented-Programming, Procedural Programming, Concurrency, Multithreading

Dev Ops Source Control, Build Tools, Dependency Management, CI/CD

Databases SQL, AWS Data Catalog, Apache Parquet, DuckDB

Libraries and Ecosystems React, Amplify, AWS, SageMaker, PyTorch Lightning, NumPy, Pandas, scikit-learn, Transformers, Transformers Reinforcement Library, Matplotlib, Plotly, Dash